**STAT2004/2904/7004 – Assignment 3**
**Due: 09 October 2026 @ 4pm**

**STAT2004/7004**: *Complete Exercises 1–4 for a maximum of 40 marks and a total of 10%.*

**STAT2904**: *Complete Exercises 1–5 for a maximum of 50 marks and a total of 10%.*

**All students**: *Exercise 6 is a bonus question. A complete and correct solution of this question earns you an extra 1% for this assignment.*

*Note that some questions involve* ***interpretation and communication of results in the form of an audio recording*** *which you upload onto Blackboard as an audio file.*

**Question 1.** **[7 marks]**

Measurements from a physics lab machine follow a normal distribution with some mean $\mu$ and some variance $\sigma^2$. To investigate the precision of this machine, 18 independent measurements, $X_1, X_2, \ldots, X_{18}$, are taken from this machine at random times over a period of three days. Based on the value of the sample variance

$$S^2 = \frac{1}{17}\sum_{i=1}^{18}(X_i - \bar{X})^2,$$

two competing confidence intervals for the unknown variance $\sigma^2$ are proposed:

$$\begin{aligned} &\text{Ross's interval:} && (0.563\,S^2,\ 2.247\,S^2)\ , \\ &\text{Alan's interval:} && (0.598\,S^2,\ 2.653\,S^2)\ . \end{aligned}$$

(a) (*1 mark*) State the distribution of the transformed random variable, $17\,S^2/\sigma^2$.

(b) (*4 marks*) Using your result from part (a), or otherwise, show that both intervals will contain the unknown $\sigma^2$ with 95% confidence.

(c) (*2 marks*) Which of these two 95% confidence intervals is "better"? Briefly justify your choice.

(a) $17\,S^2/\sigma^2 \sim \chi^2_{17}$

(b) We can start from the desired probability statement,

$$\begin{aligned} 0.95 &= P(a \le \chi^2_{17} \le b) \\ &= P(a \le 17S^2/\sigma^2 \le b) \\ &= P(17S^2/a \ge \sigma^2 \ge 17S^2/b)\ . \end{aligned}$$

For Ross' interval, this means $17/b = 0.563$ and $17/a = 2.247$, which means $b = 30.196$ and $a = 7.566$. The chance that $\chi^2_{17}$ would be between $a$ and $b$ is given by (to 3 decimal places)

```
############################################################
# P(7.566 <= chi-squared_17 <= 30.196) for Ross's interval
pchisq(30.196, 17) - pchisq(7.566, 17)
> 0.950
############################################################
```

Similarly, for Alan's interval this means $b = 28.429$ and $a = 6.408$, and the chance that $\chi^2_{17}$ would be between $a$ and $b$ is also (to 3 decimal places)

```
############################################################
# P(6.408 <= chi-squared_17 <= 28.429) for Alan's interval
pchisq(28.429, 17) - pchisq(6.408, 17)
> 0.950
############################################################
```

Hence, both Ross' and Alan's interval will contain the true $\sigma^2$ with 95% confidence.

(c) There is no correct answer here. Either of the following is reasonable: (i) If one thinks that a shorter interval (with the same level of confidence) is "better", then Ross's interval has width 2.247-0.563 = 1.684, which is narrower than Alan's interval. (ii) Both intervals have the same chance of capturing the true value of $\sigma^2$. For any given dataset, there is no guarantee that either interval will capture the true $\sigma^2$, so there is no "better" interval.

**Question 2 (Prior and posterior beliefs)** **[11 marks]**

Let's re-consider the Skittles versus M&Ms example from the lectures. Suppose that our prior belief about the underlying true proportion $p$ of Skittles is characterized by

$$f(p) \propto (1-p)^7 , \quad 0 \leq p \leq 1 .$$

(a) *(1 mark)* Graph this prior belief $f(p)$ as a function of $p$ across the range $0 \leq p \leq 1$.

(b) *(1 mark)* Qualitatively describe our prior belief. In particular, around what values of $p$ is our prior belief the strongest, and around what values of $p$ is our prior belief the the weakest?

(c) *(2 marks)* After carrying out our experiment, we observed $x = 23$ Skittles out of $n = 47$ samples. Show how our prior belief should be updated after observing these data by computing the posterior belief.

(d) *(1 mark)* Plot the posterior belief from part (c) over the range $0 \leq p \leq 1$.

(e) *(1 mark)* Qualitatively describe how our belief about $p$ has changed after observing the data.

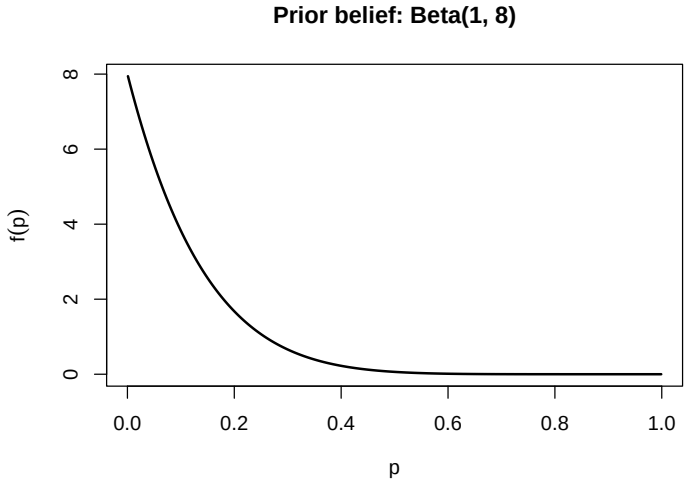
(f) *(3 mark)* Construct and interpret a 95% credible interval for the underlying true proportion $p$ based on our posterior belief.

(g) *(2 marks)* By comparing all priors and posteriors that we covered in class, tutorials, and this assignment question, comment on:

(i) the effect of the prior belief on the posterior belief ;

(ii) the effect of the data on the posterior belief.

(a) $f(p) \propto (1-p)^7$ is a Beta$(1, 8)$ density, $f(p) = 8(1-p)^7$. ✓




```
############################################################
# plot the Beta(1, 8) prior over a grid of p values
p <- seq(0.001, 0.999, length.out = 1000)
prior <- dbeta(p, 1, 8)

plot(p, prior, type = "l", lwd = 2,
     xlab = expression(p), ylab = expression(f(p)),
     main = "Prior belief: Beta(1, 8)")
############################################################
```

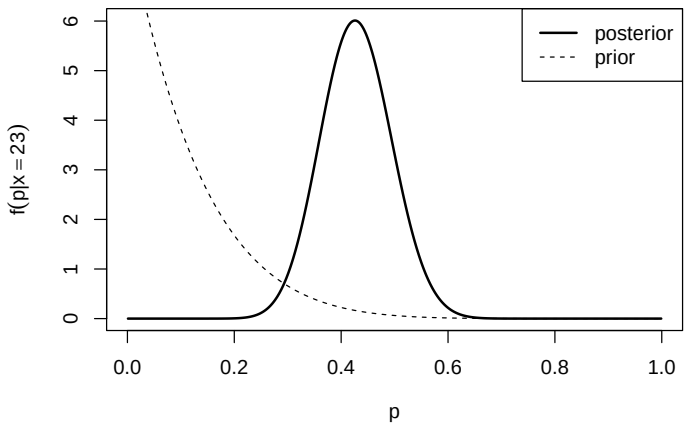
(b) Our belief is strongest near $p = 0$ and weakest near $p = 1$: we expect Skittles to be a small proportion (83% of the prior is on $p < 0.2$). ✓

(c) With $X \mid p \sim \text{Binomial}(47, p)$, the likelihood is $f(x = 23 \mid p) \propto p^{23}(1-p)^{24}$. ✓ So

$$f(p \mid x = 23) \propto p^{23}(1-p)^{24} \cdot (1-p)^{7} = p^{24-1}(1-p)^{32-1} \ ,$$

i.e. the posterior is $\text{Beta}(24, 32)$. ✓

(d) ✓



```
############################################################
# plot the Beta(24, 32) posterior, with the prior dashed
posterior <- dbeta(p, 24, 32)

plot(p, posterior, type = "l", lwd = 2,
     xlab = expression(p), ylab = expression(f(p * "|" * x == 23)),
     main = "Posterior belief: Beta(24, 32)")
lines(p, prior, lty = 2)
legend("topright", legend = c("posterior", "prior"),
       lty = c(1, 2), lwd = c(2, 1))
############################################################
```

(e) Our belief has moved from near 0 to around $p \approx 0.43$, close to the sample proportion $23/47 \approx 0.49$, and is much more concentrated. The prior still pulls it slightly below 0.49. ✓

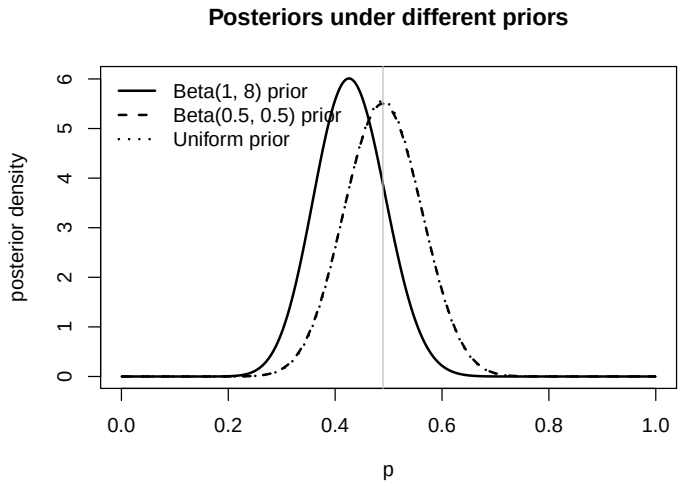
(f) Take the 2.5% and 97.5% quantiles of the posterior (any split adding to 5% is fine): ✓

```
############################################################
# 2.5% and 97.5% quantiles of the Beta(24, 32) posterior
qbeta(c(0.025, 0.975), 24, 32)
> 0.3030369 0.5589416
############################################################
```

The 95% credible interval is (0.303, 0.559). ✓ Given our prior and the data, there is a 95% probability that the true proportion of Skittles is between 0.303 and 0.559. ✓

(g) Posteriors from the same data under different priors:

| Prior | Posterior | Posterior mean | 95% credible interval |
|---|---|---|---|
| Uniform(0,1) = Beta(1, 1) | Beta(24, 25) | 0.490 | (0.352, 0.628) |
| Beta(0.5, 0.5) (tutorial) | Beta(23.5, 24.5) | 0.490 | (0.351, 0.629) |
| Beta(1, 8) (this question) | Beta(24, 32) | 0.429 | (0.303, 0.559) |




```
############################################################
# compare the posteriors under each prior
```

```
plot(p, dbeta(p, 24, 32), type = "l", lwd = 2,
     xlab = expression(p), ylab = "posterior density",
     main = "Posteriors under different priors")
lines(p, dbeta(p, 23.5, 24.5), lwd = 2, lty = 2)
lines(p, dbeta(p, 24, 25), lwd = 2, lty = 3)
abline(v = 23/47, col = "grey")   # sample proportion
############################################################
```

(i) A strongly opinionated prior has a stronger effect on the posterior. The weak Beta$(0.5, 0.5)$ and uniform priors give almost identical posteriors centred at 0.49, while the Beta$(1, 8)$ prior pulls the posterior down to 0.43. ✓

(ii) Provided the prior places non-zero belief everywhere, the data pull the posterior towards the sample proportion, whatever the prior. As $n$ grows, the effect of the prior disappears. ✓

**Question 3 (Confidence interval for a bound parameter)** **[11 marks]**

Let $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(-\theta, \theta)$, where $\theta \geq 0$ is an unknown parameter.

(a) (*2 marks*) State the distribution of $U_1 = |X_1|/\theta$.

(b) (*2 marks*) Show that the MLE of $\theta$ is given by $\hat{\theta} = \max\{|X_1|, |X_2|, \ldots, |X_n|\}$.

(c) (*2 marks*) Using part (a), or otherwise, show that $U_{(n)} = \hat{\theta}/\theta$ has cumulative distribution given by

$$P\big(U_{(n)} \leq u\big) = \begin{cases} 0\,, & u < 0 \\ u^n\,, & 0 \leq u \leq 1 \\ 1\,, & u > 1 \end{cases}$$

(d) (*1 mark*) Is $U_{(n)}$ a pivot quantity for $\theta$? Briefly explain your answer.

(e) (*1 mark*) Using part (c), or otherwise, find a pair of critical values $a$ and $b$ such that $P(a \leq U_{(n)} \leq b) = 95\%$.

(f) (*2 marks*) A sample of $n = 7$ observations is taken from this distribution, giving the following data values:

1.421 –1.197 0.476 1.106 –2.456 1.865 –0.581

Using parts (d) and (e), or otherwise, construct and interpret a 95% confidence interval for $\theta$ based on these data.

(g) (*1 mark*) [**Audio question**]: A STAT2004 student concludes that "*there is a 95% chance that the true $\theta$ is contained within the interval calculated in part (f)*". Do you agree with this conclusion? Briefly explain why, or why not.

(a) The distribution of $|X_1|$ is Uniform$(0, \theta)$ and so the distribution of $U_1 = |X_1|/\theta$ is Uniform$(0, 1)$. ✓

(b) The joint likelihood is

$$L(\theta \mid \mathbf{X}) = \begin{cases} (1/2\theta)^n \ , & -\theta \leq \text{all } X_i \leq \theta \ , \\ 0 \ , & \text{otherwise} \end{cases}$$
$$= \begin{cases} (1/2\theta)^n \ , & \theta \geq \max_i |X_i| \ , \\ 0 \ , & \text{otherwise} \end{cases} \quad \checkmark$$

This function is monotonically decreasing in $\theta$, so that the smaller $\theta$ is the larger the likelihood is. However, $\theta$ cannot be less than $\max_i |X_i|$ because the function switches to 0. Hence, the likelihood is maximized at $\hat{\theta} = \max_i |X_i|$. ✓

(c)

$$\begin{aligned} P\big(U_{(n)} \leq u\big) &= P(\max\{|X_1|/\theta, |X_2|/\theta, \ldots, |X_n|/\theta\} \leq u) \\ &= P(\text{all } |X_1|/\theta, |X_2|/\theta, \ldots, |X_n|/\theta \leq u) \quad \checkmark \\ &= P(|X_1|/\theta \leq u)\, P(|X_2|/\theta \leq u) \cdots P(|X_n|/\theta \leq u) \quad \checkmark \\ &= P(|X_1|/\theta \leq u)^n \qquad \text{by iid} \\ &= \begin{cases} 0 \ , & \text{if } u < 0 \ , \\ u^n \ , & \text{if } 0 \leq u \leq 1 \ , \\ 1 \ , & \text{if } u > 1 \ , \end{cases} \qquad \text{from part (a) that each } |X_1|/\theta \sim U(0,1) \end{aligned}$$

(d) Yes, the distribution of $U_{(n)}$ does not depend on $\theta$ so it is a pivot quantity. ✓

(e) There is no unique choice for $a$ and $b$. The most common approach is to choose "equal tail probabilities", so that $a$ and $b$ are given by:

$$0.025 = P\big(U_{(n)} \leq a\big) = a^n \quad \Leftrightarrow \quad a = (0.025)^{1/n} \ ,$$
$$0.975 = P\big(U_{(n)} \leq b\big) = b^n \quad \Leftrightarrow \quad b = (0.975)^{1/n} \ , \quad \checkmark$$

using the result from part (c).

(f) For $n = 7$, we found from part (e) that setting $a = (0.025)^{1/7} = 0.590$ and $b = (0.975)^{1/7} = 0.996$ gives us the probability statement

$$\begin{aligned} 0.95 &= P\big(0.590 \leq U_{(7)} \leq 0.996\big) \\ &= P\big(0.590 \leq |X|_{(7)}/\theta \leq 0.996\big) \qquad \text{(pivot step)} \\ &= P\left(\frac{|X|_{(7)}}{0.590} \geq \theta \geq \frac{|X|_{(7)}}{0.996}\right) \end{aligned}$$

Given the observed dataset, the sample absolute maximum is $|x|_{(7)} = 2.456$ and so a numerical 95% confidence interval is evaluated as

$$\left(\frac{2.456}{0.996}, \frac{2.456}{0.590}\right) = (2.465,\ 4.160) \ . \quad \checkmark$$

We can be 95% confident that the true upper bound $\theta$ is between 2.465 and 4.160. ✓

(g) [**Audio question**]: No, this is incorrect. There is 95% chance that this confidence interval procedure works (across all possible samples) but once we actually collect a sample of data and computed the 95% CI of $(2.465, 4.160)$ as in part (f) then there is no more chance element. ✓

**Question 4 (Hypothesis testing for a failure probability)** **[11 marks]**

Suppose that a current industry-standard engineering model predicts that a certain type of concrete block will have a 40% chance of failure when subject to a two-tonne load. A civil engineer wants to test the accuracy of this claim by obtaining 50 independent samples of such blocks, subjecting each to a two-tonne load and then recording the total number $X$ of blocks that broke. She decides that she will reject the claim $H_0 : p = 0.4$ in favour of $H_1 : p \neq 0.4$ if $X \leq 15$ or $X \geq 26$.

(a) (*1 mark*) What is the significance level $\alpha$ of this test?

(b) (*2 marks*) Graph the power of this test as a function of $p$.

The civil engineer proposes an alternative model for predicting the chance of failure of concrete blocks. Under the alternative model, the probability of failure for this type of concrete block is predicted to be 50%. She wants to test whether her new model is more accurate than the current industry standard.

(c) (*3 marks*) Construct a likelihood ratio test (LRT) of $H_0 : p = 0.4$ versus $H_1 : p = 0.5$ based on the number of failures $X$ from 50 trials.

(d) (*1 mark*) Show that the LRT is equivalent to the test with rejection region $\{X \geq c\}$ for some cutoff $c$.

(e) (*1 mark*) Changes in industry standards should not be taken lightly, so a significance level of 1% is often used in engineering instead of 5%. Find the cutoff value $c$ for carrying out this test at the 1% significance level.

(f) (*1 mark*) If the alternative model is actually correct, what is the power of this test?

(g) (*2 marks*) What sample size is needed to increase the power of this test to 90%, while keeping the level of the test at 1%?

(a) Under $H_0 : p = 0.4$, the total number $X$ of broken blocks has a $\text{Bin}(n = 50, p = 0.4)$ distribution. The level of the test is therefore given by

$$\alpha = P(X \leq 15) + P(X \geq 26) \ , \qquad \text{where } X \sim \text{Bin}(50, 0.4)$$

This should be done using software. For example, using `R` we find that the level is $\alpha = 15.3\%$ ✓

```
############################################################
# compute P(X <= 15) for X ~ Bin(50, 0.4)
pbinom(15, 50, 0.4)
> 0.09550171

# compute P(X >= 26) for X ~ Bin(50, 0.4)
1-pbinom(25, 50, 0.4)
> 0.05734376

# P(X <= 15) + P(X >= 26) for X ~ Bin(50, 0.4)
pbinom(15, 50, 0.4) + 1-pbinom(25,50,0.4)
> 0.1528455
############################################################
```

Some students may have used the Normal approximation to the Binomial distribution, which is fine. But at some point, they have to convert $z$-values into a probabilities, which would require either looking up a Normal table or using software anyway. So might as well work directly with the $\text{Bin}(50, 0.4)$ distribution on a computer and avoid any approximation errors.

(b) The power of a test is the probability of a correct rejection under the alternative. For each alternative value $p_1$, the power is therefore given by

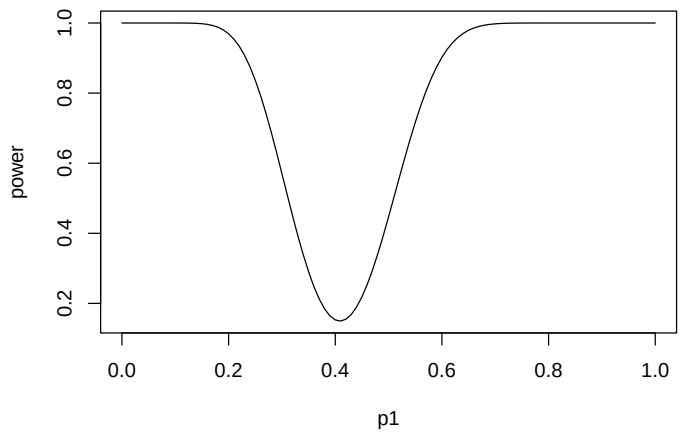
$$\text{power}(p_1) = P(X \leq 15) + P(X \geq 26) \ , \qquad \text{where } X \sim \text{Bin}(50, p_1) \quad \checkmark$$

This function can be evaluated at a grid of values for $p_1$ between 0 and 1, giving the power curve below: ✓

```
#############################################################
# set up a grid of p1 values from 0 to 1 in steps of 0.01
p1 = seq(0, 1, by=0.01)

# compute the power at each p1 value
power = pbinom(15, 50, p1) + 1-pbinom(25,50,p1)

# plot the power curve
plot(p1, power, type="l")
#############################################################
```




(c) The likelihood function for the $p$ given the data $X$ is

$$L(p \mid X) = \binom{50}{X} p^X (1-p)^{50-X} \ , \quad \checkmark$$

so that the likelihood ratio test statistic for testing $H_0 : p = 0.4$ versus $H_1 : p = 0.5$ is

$$\frac{L(0.4 \mid X)}{L(0.5 \mid X)} = \frac{0.4^X \, 0.6^{50-X}}{0.5^X \, 0.5^{50-X}} \quad \checkmark$$

$$= \left(\frac{0.6}{0.5}\right)^{50} \left(\frac{0.4}{0.6}\right)^X = 1.2^{50} \left(\frac{2}{3}\right)^X \ . \quad \checkmark$$

(d) Since $2/3 < 1$, small values of the LRT statistic correspond to large values of $X$, and so the LRT is equivalent to the test with rejection region $\{X \geq c\}$ for some cutoff $c$. ✓

(e) We want to find $c$ such that

$$0.01 \geq P(X \geq c) \ , \qquad \text{where } X \sim \text{Bin}(50, 0.4) \text{ under the null.}$$

Using software, we find that $c$ must be 29 (giving a level of 0.76%): ✓

```
############################################################
# find the cutoff for which there is at least 99% to the left
qbinom(0.99, 50, 0.4)
> 28

# check the level P(X >= 29) for X ~ Bin(50, 0.4)
1-pbinom(28, 50, 0.4)
> 0.007617426
############################################################
```

(f) The decision rule is $\{X \geq 29\}$. If the alternative value of $p = 0.5$ is true, the power of this test is

$$\text{power} = P(X \geq 29) \ , \qquad \text{where } X \sim \text{Bin}(50, 0.5) \text{ under the alternative.}$$

Using software, we find the power to be 16.1%: ✓

```
############################################################
# find the power of this test
1-pbinom(28, 50, 0.5)
> 0.1611182
############################################################
```

(g) As we change sample sizes, we need to change the cutoff $c$ to maintain a Type I error of no more than 1%. (It is nonsensical to keep testing whether the total number of counts $\geq 29$ if the sample size increases to 100 or 200...). To do this, we simply modify the code from parts (e) and (f) and keep increasing the sample size $n$ until 90% power is achieved:

```
############################################################
# start with sample size n = 50
n = 50
# find c-1 for which there is at least 99% to the left
c.minus.1 = qbinom(0.99, n, 0.4)

# find the power of this test P(X >= c) under alternative
power = 1 - pbinom(c.minus.1, n, 0.5)
# keep increasing sample size until power hits 90%
while(power < 0.9){
  n = n+1
  c.minus.1 = qbinom(0.99, n, 0.4)
  power = 1 - pbinom(c.minus.1, n, 0.5)
}
# what is the sample size we need?
n
```

```
> 322
# what is the cutoff c-1 we need to maintain 1% Type I error?
c.minus.1
> 149
# what is the power of this test?
power
> 0.9000703
############################################################
```

So the sample size we need is $n = 322$ concrete blocks ✓ and the decision rule that rejects $H_0 : p = 0.4$ in favour of $H_1 : p = 0.5$ if {number of failures $X \geq 150$} has power 90.0%. ✓

Because the alternative $p = 0.5$ is close to the null $p = 0.4$, a large sample size is needed to detect this difference with high power while protecting the Type I error at 1%.

**Question 5 (STAT2904 only)** **[10 marks]**

Let $X_1, X_2, \ldots, X_n$ be an iid sample from a continuous distribution with pdf

$$f_\theta(x) = \frac{2x}{\theta^2} , \qquad 0 \leq x \leq \theta ,$$

where $\theta > 0$ is an unknown parameter.

(a) (*2 marks*) Derive the MLE $\hat{\theta}$ of $\theta$.

(b) (*2 marks*) Find the cdf of $\hat{\theta}$.

(c) (*1 mark*) Using part (b), construct a pivot for the parameter $\theta$ via the cumulative probability transform method. State the distribution of your pivot variable.

(d) (*2 marks*) Using your pivot from part (c), show that *any* interval of the form,

$$\left( \frac{X_{(n)}}{(a + 0.95)^{1/2n}}, \frac{X_{(n)}}{a^{1/2n}} \right) , \quad 0 \leq a \leq 0.05 ,$$

defines a 95% confidence interval for $\theta$.

(e) (*2 marks*) Find the value $a$ that minimises the length of the interval from part (d).

(f) (*1 mark*) Enumerate the shortest 95% confidence interval for $\theta$ based on the following realisations from $f_\theta(x)$: 1.227 3.829 4.813 4.874 5.142.

(a) The likelihood is

$$L(\theta \mid \mathbf{X}) = \begin{cases} \dfrac{2^n}{\theta^{2n}} \displaystyle\prod_{i=1}^{n} X_i , & \text{if } 0 < \text{all } X_i < \theta \\ 0 , & \text{otherwise} \end{cases}$$

We see that the smaller $\theta$ is, the larger the likelihood will be – HOWEVER, $\theta$ cannot be smaller than $X_{(n)} = \max\{X_1, X_2, \ldots, X_n\}$ otherwise the likelihood switches to 0. Hence, the MLE is $\hat{\theta} = X_{(n)}$. ✓✓

(b)

$$\begin{aligned} P(\hat{\theta} \le m) &= P(\text{all } X_i \le m) \\ &= \prod_{i=1}^{n} P(X_i \le m) \\ &= \left( \int_0^m \frac{2x}{\theta^2} \, \mathrm{d}x \right)^n \\ &= \left( \frac{m^2}{\theta^2} \right)^n \end{aligned}$$

So the cdf of $\hat{\theta}$ is given by $F(m) = (m/\theta)^{2n}$. ✓✓

(c) Because $\hat{\theta}$ has a continuous distribution, the cdf transformation is always a pivot:

$$U = F(\hat{\theta}) = \left( \hat{\theta}/\theta \right)^{2n}$$

which has a Uniform(0,1) distribution. ✓

(d) Starting from the known probability statement about the uniform distribution,

$$\begin{aligned} 0.95 &= P(a \le U \le a + 0.95) \ , \quad \text{for any } 0 \le a \le 0.05 \\ &= P\Big( a \le (\hat{\theta}/\theta)^{2n} \le a + 0.95 \Big) \\ &= P\Big( a^{1/2n} \le \hat{\theta}/\theta \le (a + 0.95)^{1/2n} \Big) \\ &= P\left( \frac{\hat{\theta}}{(a + 0.95)^{1/2n}} \le \theta \le \frac{\hat{\theta}}{a^{1/2n}} \right) , \end{aligned}$$

we have a 95% confidence interval for $\theta$ as

$$\left( \frac{X_{(n)}}{(a + 0.95)^{1/2n}}, \frac{X_{(n)}}{a^{1/2n}} \right) , \quad \text{for any } 0 \le a \le 0.05. \quad ✓✓$$

(e) The length of the interval is

$$\left[ \frac{\hat{\theta}}{a^{1/2n}} - \frac{\hat{\theta}}{(a + 0.95)^{1/2n}} \right] \quad ✓$$

This is a monotonically decreasing function in $a$ over the range $a \in [0, 0.05]$, and setting $a = 0.05$ ✓ leads to shortest confidence interval which has the form

$$\left( \frac{X_{(n)}}{1}, \frac{X_{(n)}}{(0.05)^{1/2n}} \right) .$$

(f) For the $n = 5$ observations, the maximum was $x_{(n)} = 5.142$ and so the shortest numerical 95% confidence interval from this dataset is given by

$$\left( 5.142, \frac{5.142}{(0.05)^{1/10}} \right) = \left( 5.142, \frac{5.142}{0.741} \right) = (5.142, 6.938) \ . \quad ✓$$

We can be 95% confident that the true $\theta$ that generated this dataset is between 5.142 and 6.938.

**Question 6 (Bonus Question for all students)** **[4 marks]**

Let $X_1, X_2, \ldots, X_{17}$ be a random sample of size $n = 17$ coming from some arbitrary continuous distribution with unknown population median, $m$.

(a) Show that the interval,

$$\left[X_{(5)},\, X_{(13)}\right] ,$$

defines an (approximate) 95% confidence interval for the population median, $m$. Here, $X_{(5)}$ denotes the fifth smallest observation, and $X_{(13)}$ denotes the fifth largest observation.

Confidence intervals such as these for the population median are intriguing because they depend only on order statistics and do not require any actual model for the underlying distribution of the data!

(b) Suppose the underlying population distribution is actually $t_2$, which has a median (and mean) equal to 0. Using computer simulations, or otherwise, compare the relative merits of the interval from part (a) and the classical $\bar{X} \pm t_{16}^{(0.975)} S/\sqrt{17}$ confidence interval for capturing the true mean/median of 0.

You may use the following `R` code to simulate $n = 17$ observations from a $t_2$ distribution:

```
> rt(17, 2)
```

(a) The proposed interval will NOT contain the true median $m$ if fewer than 5 observations fall below the median $m$ OR if fewer than 5 fall above the median $m$. ✓ Because the chance of any observation being above or below the median is 0.5 (by definition of the median) ✓, then the chance that the proposed interval will NOT contain the true median is (using `R` syntax)

```
############################################################
2*pbinom(4, 17, 0.5)
> 0.04904175
# or equivalently #
pbinom(4, 17, 0.5) + 1-pbinom(12, 17, 0.5)
> 0.04904175
############################################################
```

Hence, the chance that this interval procedure contains the true median $m$ is $1 - 0.049 = 0.951 = 95.1\%$, and this is therefore a 95% confidence interval. ✓

(b) Simulate $B = 10{,}000$ samples of size $n = 17$ from $t_2$, compute both intervals for each sample, and compare how often each captures the true median 0 (coverage) and how wide they are:

```
############################################################
set.seed(2004)
B = 10000
n = 17
order.ci = matrix(NA, B, 2)
t.ci = matrix(NA, B, 2)
for(b in 1:B){
```

```
  x = rt(n, 2)
  # order statistic interval [X(5), X(13)]
  order.ci[b, ] = sort(x)[c(5, 13)]
  # classical t interval
  t.ci[b, ] = mean(x) + c(-1, 1) * qt(0.975, n-1) * sd(x)/sqrt(n)
}
# coverage: proportion of intervals containing 0
mean(order.ci[,1] <= 0 & 0 <= order.ci[,2])
> 0.9525
mean(t.ci[,1] <= 0 & 0 <= t.ci[,2])
> 0.9634

# interval widths
summary(order.ci[,2] - order.ci[,1])
>   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
> 0.2892  1.1830  1.4969  1.5632  1.8751  5.9480
summary(t.ci[,2] - t.ci[,1])
>   Min. 1st Qu.  Median    Mean 3rd Qu.     Max.
> 0.5196  1.4269  1.8445  2.4368  2.5305 840.6009
############################################################
```

Both intervals capture 0 about 95% of the time. The order statistic interval matches its exact coverage of 95.1% from part (a), as it should for any continuous distribution. However, the $t_2$ distribution has heavy tails (infinite variance), so a single extreme observation inflates $\bar{X}$ and $S$, and the $t$ interval can become very wide. So for heavy-tailed data the order statistic interval gives the correct coverage with tighter intervals.